

Plasma LDL and HDL cholesterol and oxidized LDL concentrations are altered in normo- and hypercholesterolemic humans after intake of different leve...

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Cocoa powder is rich in polyphenols, such as catechins and procyanidins, and has been shown in a variety of subject models to inhibit oxidized LDL and atherogenesis. Our study evaluated plasma LDL cholesterol and oxidized LDL concentrations following the intake of different levels of cocoa powder (13, 19.5, and 26 g/d) in normocholesterolemic and mildly hypercholesterolemic humans. In this comparative, double-blind study, we examined 160 subjects who ingested either cocoa powder containing low-polyphenolic compounds (placebo-cocoa group) or 3 levels of cocoa powder containing high-polyphenolic compounds (13, 19.5, and 26 g/d for low-, middle-, and high-cocoa groups, respectively) for 4 wk. The test powders were consumed as a beverage after the addition of hot water, twice each day. Blood samples were collected at baseline and 4 wk after intake of the test beverages for the measurement of plasma lipids. Plasma oxidized LDL concentrations decreased in the low-, middle-, and high-cocoa groups compared with baseline. A stratified analysis was performed on 131 subjects who had a LDL cholesterol concentrations of ≥ 3.23 mmol/L at baseline. In these subjects, plasma LDL cholesterol, oxidized LDL, and apo B concentrations decreased, and the plasma HDL cholesterol concentration increased, relative to baseline in the low-, middle-, and high-cocoa groups. The results suggest that polyphenolic substances derived from cocoa powder may contribute to a reduction in LDL cholesterol, an elevation in HDL cholesterol, and the suppression of oxidized LDL.